

ELEC 221 Lecture 05

Impulse response stability and causality; CT Fourier series

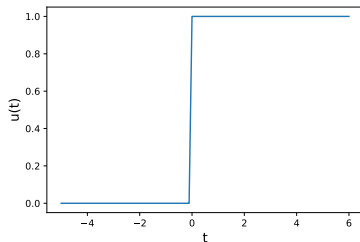
Thursday 24 September 2026

Announcements

- Tutorial Assignment 1 due tomorrow night at 23:59
- Quiz 3 Tuesday
- Tutorial 2 on Monday: practice problems

Last time

We defined the CT unit step, $u(t)$



And the CT unit impulse function, $\delta(t)$:

Last time

We introduced the CT convolution integral,

If we know the impulse response of an LTI system, we can determine its response to any other signal.

Last time

We saw how properties of the impulse response are related to the system's properties as a whole.

(**Memory**) For a system to be memoryless,

(**Invertibility**) If a system with impulse response $h(t)$ is invertible, there exists an *inverse system* with impulse response $h_i(t)$ s.t.

(Analogous for DT case)

Today

Learning outcomes:

- Use the impulse response to determine whether a system is stable or causal
- Use the convolution integral to show that complex exponentials are eigenfunctions of LTI systems
- Define and compute the system function (frequency response) of an LTI system
- Express a periodic CT signal as a Fourier series, and compute its Fourier coefficients

Impulse response and stability

Suppose we have a signal $x(t)$ with a bounded input, $|x(t)| \leq B$.
If the system is stable, the output should be bounded.

Impulse response and stability

As long as

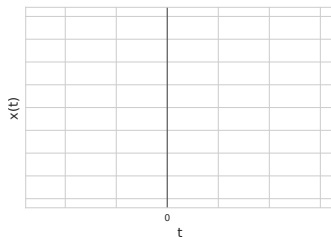
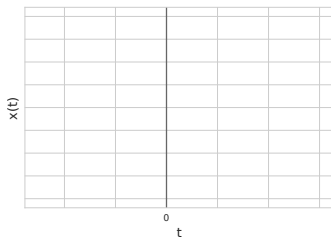
(i.e., $h(t)$ is *absolutely integrable*), the system is stable.

In DT, $h[n]$ must be *absolutely summable* to ensure stability:

Example/exercise: stability

Consider systems A and B with impulse responses

Are they stable?



Example/exercise: stability

Impulse response and causality

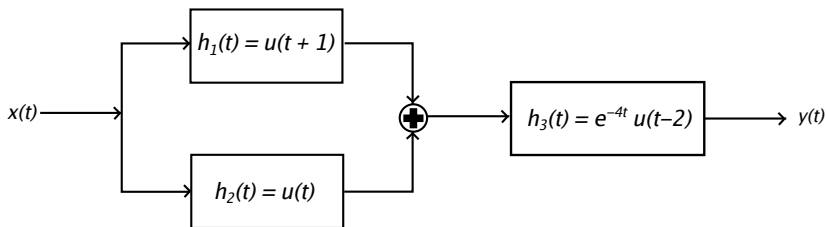
Recall definition of causal signal and consider the convolution sum:

What properties does $h[n]$ need to have for system to be causal?

(Analogous holds for CT systems)

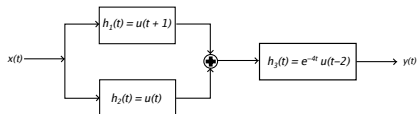
Final exercise

Consider the following combination of systems:

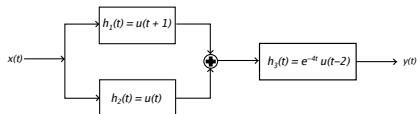


Is this system causal and/or stable?

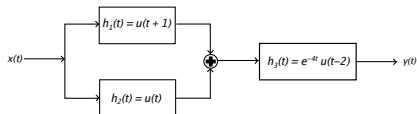
Final exercise



Final exercise



Final exercise

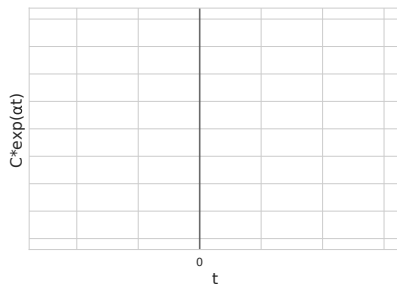
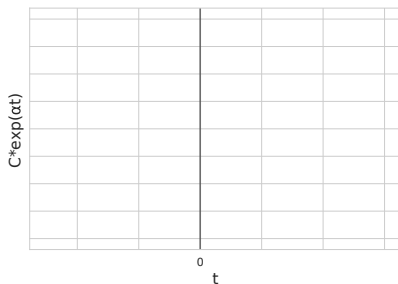


Review: complex exponential functions

Most general form:

where α can be real or complex.

Case: both C and α are real-valued.



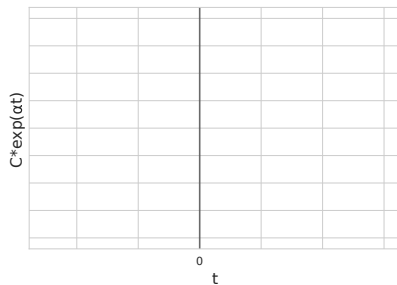
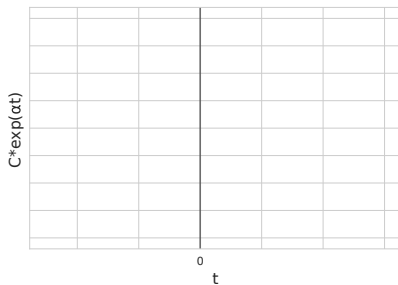
Review: complex exponential functions

Case: α is complex. Most generally, we can write

Euler's relation allows us to write

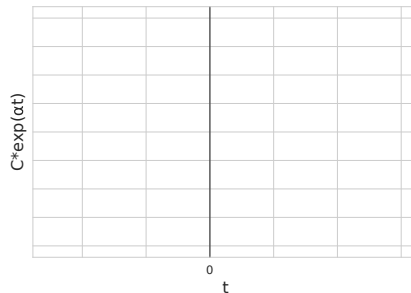
Review: complex exponential functions

Case: $\alpha = j\omega$. Then,



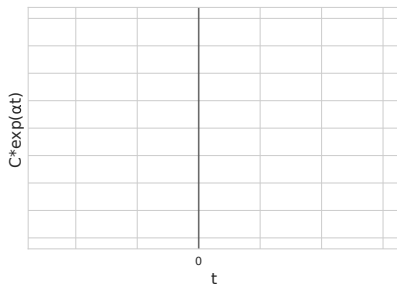
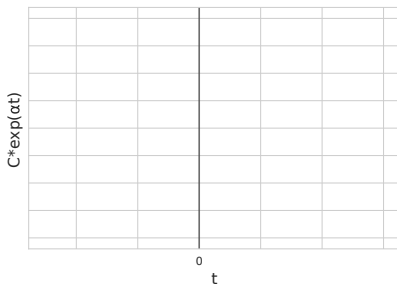
Review: complex exponential functions

$x(t) = Ce^{j\omega t}$ is periodic:



Review: complex exponential functions

Case: $\alpha = r + j\omega$. Then,



LTI systems and complex exponential functions

Recall the convolution integral:

What happens when $x(t)$ is a complex exponential signal?

LTI systems and complex exponential functions

Write $x(t) = e^{st}$. Then:

LTI systems and complex exponential functions

To summarize:

Complex exponentials are **eigenfunctions** of LTI systems.

$H(s)$ is the **system function**.

...so what?

Recall that for LTI systems,

If all $x_i(t)$ are complex exponential signals, and

then

The response is a superposition **of those same signals**, scaled by the system function.

System functions

Consider the limited set of signals¹:

$x(t)$ has frequency ω and period $T = 2\pi/\omega$.

When such signals are input into an LTI system, the system function, $H(s) = H(j\omega)$, is called the *frequency response*.

¹We will see the general case at the end of the course.

Exercise: frequency response

Suppose we have access to a system with impulse response

$$h(t) = \frac{1}{2}\delta(t - 1)$$

1. What is the frequency response, $H(j\omega)$, of the system?

Exercise: frequency response

Suppose we have access to a system with impulse response

$$h(t) = \frac{1}{2}\delta(t - 1)$$

2. What is the output of the system given input

$$x(t) = 2e^{2j\omega t} - 3e^{5j\omega t}$$

Exercise: frequency response

Suppose we have access to a system with impulse response

$$h(t) = \frac{1}{2}\delta(t - 1)$$

3. What is the output of the system given input

$$x(t) = \frac{2}{3}\cos(3\omega t)$$

The Fourier series

How can we express arbitrary signals using complex exponentials?

The Fourier series

Consider

It also has an infinite number of cousins called *harmonics*.

ω is the **fundamental frequency**.

(Yes, these harmonics.)

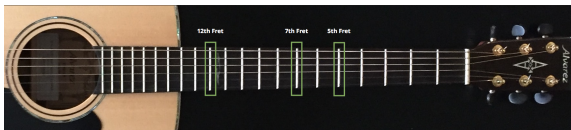


Image credit: <https://musiciantuts.com/guitar-harmonics/>

The Fourier series

We can create a superposition of all harmonics,

This signal is also periodic with period $T = 2\pi/\omega$.

Given $x(t)$ with period T^2 , we can write it as a **Fourier series**.

The c_k (complex numbers) are its **Fourier coefficients**.

²There are some additional conditions; we will cover this next class

The Fourier series

Usually dealing with $x(t)$ that is always *real*.

This means .

We can leverage this to express $x(t)$ in a different way.

The Fourier series

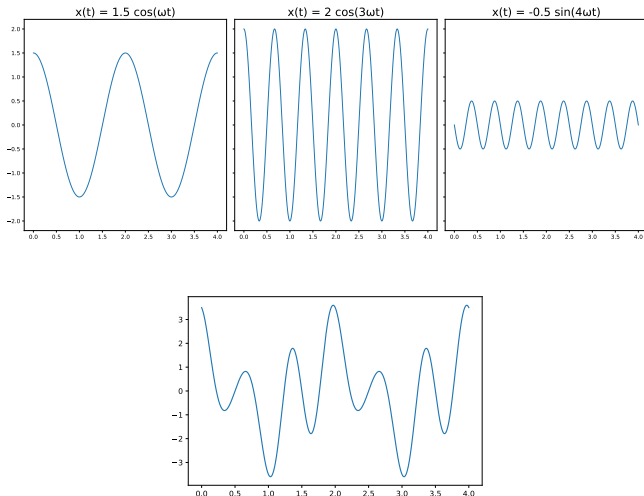
Apply Euler's relation:

With a bit of work (i.e., in assignment 1!), you can find

where a_0 , a_k , b_k are expressed in terms of the c_k .

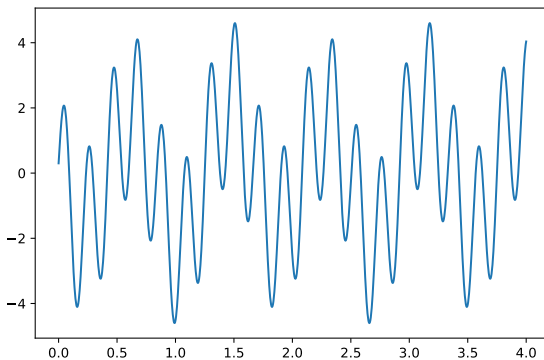
Note: sometimes a_0 is written $a_0/2$ for convenience

Example



Going backwards

What if we are given a signal like this:



Can we determine the c_k ?

Evaluating Fourier coefficients

The $e^{jk\omega t}$ are **basis functions** and have **orthogonality** relations w.r.t. integration.

Example: let's compute c_m .

Multiply on both sides by the conjugate of the basis function:

Evaluating Fourier coefficients

$$e^{-jm\omega t}x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t} e^{-jm\omega t}$$

Integrate over a period:

Evaluating Fourier coefficients

Evaluate the integral

Case 1: $k = m$

Case 2: $k \neq m$

Evaluating Fourier coefficients

From here:

$$\frac{1}{T} \int_0^T e^{-jm\omega t} x(t) dt = \sum_{k=-\infty}^{\infty} c_k \cdot \frac{1}{T} \int_0^T e^{j(k-m)\omega t} dt$$

Only the $k = m$ term survives so

Evaluating Fourier coefficients

Since we integrate over one period, write

Fourier coefficients tell us *how much* each harmonic contributes to the total signal.

Note that c_0 is a constant offset:

(Similar techniques can be used to derive a_k and b_k for the sin and cos representation. Try it!)

Recap: key expressions

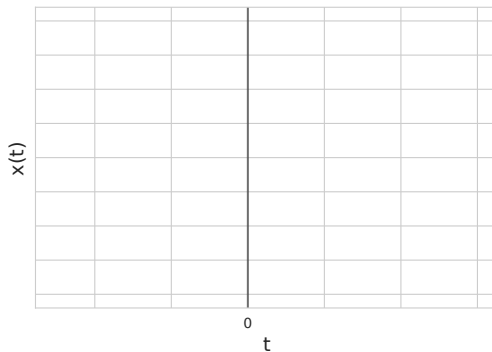
Fourier synthesis equation:

Fourier analysis equation:

Exercise

What is the Fourier series of

Start with a plot, and determine T and ω .



Exercise

Exercise

For next time

Content:

- Dirichlet conditions and the Gibbs phenomenon
- Properties of Fourier series and Fourier coefficients

Action items:

- ① Assignment 1 is due tonight
- ② Tuesday quiz on this week's material

Recommended reading:

- From today's class: Oppenheim 1.3, 3.0-3.3
- Suggested problems: 1.6, 1.8, 1.10, 3.1, 3.3, 3.4, 3.13, 3.17, 3.22a,c
- For next class: Oppenheim 3.4-3.5